Federated Full-Envelope LPV Identification from Incomplete Local Coverage

***Abstract*—Vehicles may supply calibration data from narrow speed regions while the fleet collectively covers a wider interval. Pooling can recover missing scheduling information but mix incompatible dynamics. We study federated linear-parameter-varying (LPV) identification under these joint limitations. Clients exchange additive regression statistics and assignment scores to learn shared group maps. Across ten nonlinear vehicle fleets with family labels withheld, grouped fitting reduces mean paired full-envelope matrix error by 56.20% relative to development-tuned Smooth Local and 23.74% relative to Global. Mean error at locally unobserved speeds is 3.527%, versus 9.458% and 4.388%, respectively. The distributed fit reproduces centralized grouping. Benefits persist under alternative entry scaling but are not uniform across entries or clients. The benchmark uses controlled equilibrium-centered calibration with available states, exact current regressors, and noisy next-state targets. Fresh stress tests expose fleet-wide coverage failure and loss of grouping gains at lower calibration signal-to-noise. Speed-scheduled linear-quadratic-integral (LQI) tests show modest gains over Global; the tested Smooth Local prior yields infeasible responses for some clients under the frozen controller protocol.**

***Index Terms*—federated identification, LPV systems, incomplete operating coverage, heterogeneous vehicles**

I. Introduction

Vehicles with speed-dependent and heterogeneous dynamics may observe only narrow speed regions. LPV identification then requires complementary scheduling information while avoiding bias from pooling incompatible dynamics.

Basis-dependent LPV identification requires scheduling excitation [1], [2]. Local approaches identify fixed-operating-point models before constructing a scheduling-dependent map [3]. Federated learning retains records locally [4]. For heterogeneous systems, FedSysID quantifies the tradeoff between collaboration and model bias [5]. IFCA alternates client assignment and cluster-model updates [6]. Clustered SysID applies alternating assignment and estimation to linear time-invariant (LTI) systems with identical dynamics within each true cluster, providing conditional recovery and sample-complexity guarantees [7]. IC-SYSID adds incremental cluster formation and stability promotion, including a vehicle-fleet study [8]. The vehicle study in [9] uses uncertainty-aware grouping for controller design. Here the central question is whether sharing can identify an LPV map when each client lacks scheduling directions, rather than how uncertainty in locally estimated dynamics informs grouping for controller design.

Here each client's fixed speed support leaves its LPV scheduling design deficient. Repeating transitions at those speeds reduces noise but cannot identify the missing coefficient directions. Compatible pooling can supply them, conditional on group coverage, while within-group variation remains. The unresolved issue is whether assignment from partial-speed targets can produce groups that are both sufficiently compatible and collectively informative across the speed interval. With common physical-state coordinates, we specialize the established alternating assignment and estimation structure to LPV matrix regression. The cited recovery guarantees do not establish recovery for our fixed deficient scheduling supports and approximate shared maps. Our contribution is the LPV formulation and controlled evidence for this joint limitation; Proposition 1 states the conditional rank requirement, while the experiments assess its satisfaction and the remaining pooling bias. Figure 1 summarizes the procedure.

Blocked-speed validation separates representation error from missing information. Additive statistics and local scores implement grouped regression without full local coefficient maps or family labels; centralized agreement checks implementation. Paired fleet comparisons and group-rank audits assess coverage and compatibility, while fresh stress tests bound their benefit. Speed-scheduled linear-quadratic-integral (LQI) control tests downstream usability on nonlinear vehicles.

II. Identification Problem

A. Available states and incomplete speed coverage

Client i has sideslip angle β, yaw rate r, state $x = [\beta, r]^\top$, steering input u, and longitudinal speed v. The benchmark uses radians for β, u and rad/s for r. At a fixed cornering equilibrium, local perturbations satisfy approximately

$$\Delta x_{i,k+1} = A_i(v)\Delta x_{i,k} + B_i(v)\Delta u_{i,k} + e_{i,k}, \qquad (1)$$

where A_i, B_i are discrete-time matrices and $e_{i,k}$ includes nonlinear approximation error and next-state observation noise. Here $\Delta x = x - x_i^\star(v)$ and $\Delta u = u - u_i^\star(v)$ are deviations from the prescribed cornering equilibrium; the calibration workflow is specified in Section IV-A. Define $m_i(v) = [A_{11}, A_{12}, A_{21}, A_{22}, B_1, B_2]$. The learned group assignment g_i defines the approximation

$$m_i(v) \approx \phi(v)^\top \Theta_{g_i}, \qquad \Theta_g \in \mathbb{R}^{L\times 6}. \qquad (2)$$

Shared maps retain within-group approximation error. Each client provides matrix estimates at speeds in $\mathcal{V}_i \subset \mathcal{V} = [10, 30]$ m/s. "Full envelope" means this speed interval along the prescribed cornering curve, rather than an arbitrary nonlinear state/input domain.

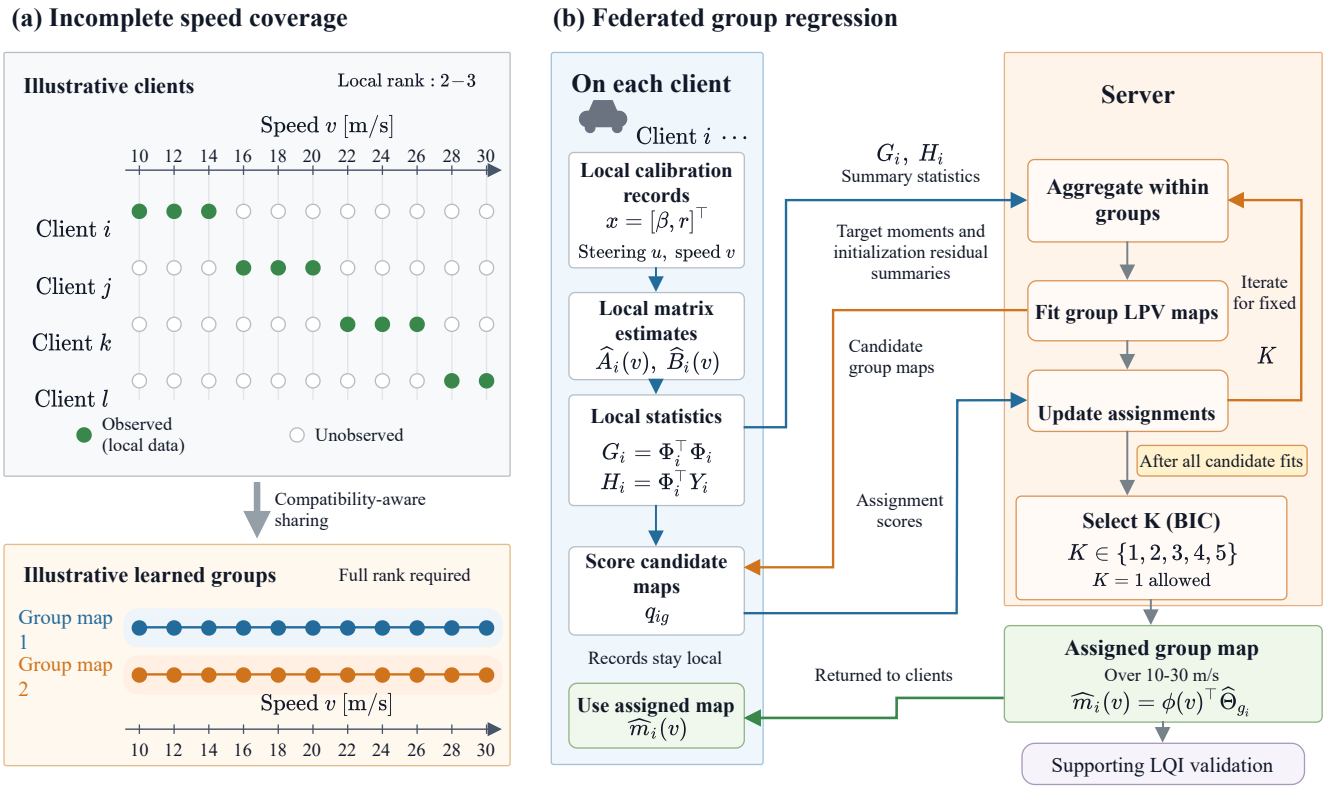


Fig. 1. Compatible federated LPV identification. (a) Illustrative clients and groups. Local speed blocks give scheduling rank two or three for a seven-function basis; sharing can supply missing directions when the pooled group design has rank seven. (b) Clients retain records and matrix targets, exchange additive statistics, residual summaries, and assignment scores, and receive group maps. Fitting and assignment alternate for each candidate K, followed by selection using the Bayesian information criterion (BIC). Available states and equilibrium-centered calibration are assumed. The envelope is 10–30 m/s along the prescribed cornering curve; LQI checks downstream usability.

B. Fixed representation and local estimation

The scheduling span contains $L = 7$ functions:

$$b(v) = [1, v^{-1}, v^{-2}, z, z^2, z^3, z^4]^\top, \quad z = (v - 20)/10. \quad (3)$$

Let the rows of B_{ref} be $b(v)^\top$ on a fixed 401-point uniform 10–30 m/s grid, with thin QR factorization $B_{\text{ref}} = QR$. The implemented row basis is $\phi(v)^\top = b(v)^\top R^{-1}$. This invertible transform changes conditioning, not the represented span. Development-only blocked-speed validation selects $L = 7$ for cornering, while the straight-line negative control retains $[1, v^{-1}, v^{-2}]$. No control comparison selects the representation.

At each observed speed, a client stacks 36 independent transitions. With rows $[\Delta x_k^\top, \Delta u_k]$ in Z and next-state rows in X_+, ordinary least squares (OLS) gives

$$\widehat{C} = \arg\min_C \|X_+ - ZC\|_F^2, \qquad [\widehat{A}, \widehat{B}] = \widehat{C}^\top. \quad (4)$$

In the primary study, current state/input regressors are exact, and only next-state targets receive noise. The resulting matrix estimates and their speeds are the group-regression targets. Vehicle parameters, exact evaluation matrices, and family labels do not enter that fitting or assignment interface.

The benchmark supplies centering equilibria from simulator parameters. Section IV-C separately perturbs current measurements and assumed centers. Estimating these centers from ordinary driving remains unvalidated. The clients contributing calibration records also receive the fitted maps. Evaluation at additional speeds tests generalization across operating conditions. Each fresh fleet repeats identification and grouping; transfer of a fixed model library to new recipients is not evaluated.

III. Federated Compatible Regression

A. Additive fitting and conditional identifiability

Let $n_i = |\mathcal{V}_i|$ denote the number of observed speeds, not the number of transitions. Client i forms $\Phi_i \in \mathbb{R}^{n_i \times L}$ and matrix-entry targets $Y_i \in \mathbb{R}^{n_i \times 6}$. It contributes $G_i = \Phi_i^\top \Phi_i$ and $H_i = \Phi_i^\top Y_i$. For a fixed partition, the group fit is

$$\widehat{\Theta}_g = \left(\sum_{i:g_i=g} G_i + (\lambda + \epsilon) I \right)^{-1} \sum_{i:g_i=g} H_i, \quad (5)$$

with $\lambda = 10^{-8}$ and numerical diagonal $\epsilon = 10^{-12}$. The same unweighted ridge objective and basis are used in centralized and federated grouped fitting. The matrix estimates have equal weight; local estimation covariance is not propagated. The sensitivity study retains this unweighted rule.

Proposition 1: For a fixed partition and exact shared-map targets $Y_i = \Phi_i \Theta_g$, the unregularized coefficients are unique

TABLE I
FEDERATED GROUP-REGRESSION PROCEDURE.

Client inputs: Local speed/matrix pairs, fixed basis, candidate $K = 1, \dots, 5$, minimum group size five.
1. Aggregate local statistics and target moments. Fit Global, broadcast it, and collect residual summaries.
2. For each K, use one initialization for $K = 1$ or twelve seeded K-means initializations otherwise.
3. Alternate group-statistic solves, map broadcasts, local score returns, and hard assignments. Stop at unchanged assignments or 30 iterations. Discard initializations with groups smaller than five.
4. Refit the final admissible partition if needed. Retain the lowest-BIC initialization for each K, then the lowest-BIC K.
Outputs: Group maps and client assignments, including $K = 1$.

if and only if the vertically stacked group design has rank L, equivalently $\operatorname{rank}(\sum_{i:g_i=g} G_i) = L$.
Justification: The aggregate Gram matrix is positive definite exactly when the stacked design has no nonzero null direction. Otherwise, adding a null-direction coefficient matrix leaves all observed predictions unchanged. Additivity also makes the centralized and distributed solutions identical under the same numerical conventions.

This statement is conditional on a partition and model class. It neither establishes correct group discovery nor removes pooling bias. Ridge gives a numerical answer in deficient groups but does not restore information.

B. Assignment and model selection

The server derives a common entry scale from additive counts, sums, and squared sums of Y_i. For entry j, let s_j be its pooled standard deviation, floored at 0.05 times the median standard deviation, and set $D = \operatorname{diag}(s_1, \dots, s_6)$. Clients return scores

$$q_{ig} = \|(Y_i - \Phi_i \widehat{\Theta}_g) D^{-1}\|_F^2, \qquad g_i \leftarrow \arg\min_g q_{ig}. \tag{6}$$

Assignment ties choose the lowest group index. The common scaling balances entries during assignment; fitting uses entry-wise ridge regression. Initial groups come from K-means on standardized six-entry client mean residuals around the Global fit. The server receives those summaries, not the local matrices themselves.

The BIC score uses the normalized assignment residuals:

$$\mathrm{BIC} = n \log\big(\max\{\mathrm{SSE}/n, 10^{-12}\}\big) + p_K \log n, \tag{7}$$

where $\mathrm{SSE} = \sum_i q_{i,g_i}$, $n = 6\sum_i n_i$, and $p_K = 6LK + K - 1$. This is a heuristic selection criterion for finite noisy matrix targets and within-group variation. It does not certify a likelihood model, global convergence, or physical-family recovery. Minimum group size does not imply full scheduling rank, so group fit and group information must be assessed separately.

C. Communication

For $L = 7$, each local Gram has 28 distinct entries and its right-hand side has 42. Target moments, initialization summaries, group tags, assignment scores, and model broadcasts also enter the accounting. Including all candidate group counts and restarts, mean label-free traffic is 2.243 MB per fleet at eight bytes per scalar. This exhaustive search is not shown to be communication-efficient. The emulator retains raw records locally, but informative statistics and scores are shared. No secure aggregation or differential privacy is implemented.

TABLE II
VEHICLE PARAMETER CENTERS USED TO GENERATE THE FLEET. ALL CLIENTS HAVE $l_f = 1.2$ M AND $l_r = 1.6$ M.

Center	m [kg]	I_z [kg m^2]	C_f	C_r
Nominal	1500	2500	80000	80000
Heavy	1725	3000	84000	80000
Handling	1425	2250	72000	88000

Stiffnesses in N/rad.

IV. BENCHMARK AND EVALUATION

A. Nonlinear vehicles and calibration records

The plant is a two-state single-track model with nonlinear tanh tires:

$$\dot{\beta} = (F_f + F_r)/(mv) - r, \tag{8}$$

$$\dot{r} = (l_f F_f - l_r F_r)/I_z, \tag{9}$$

$$F_a = \mu F_{za} \tanh(C_a \alpha_a/(\mu F_{za})), \quad a \in \{f, r\}. \tag{10}$$

Here $\alpha_f = u - \beta - l_f r/v$, $\alpha_r = -\beta + l_r r/v$, $F_{zf} = mgl_r/(l_f + l_r)$, $F_{zr} = mgl_f/(l_f + l_r)$, $g = 9.81$ m/s^2, and $\mu = 0.9$. Each fleet has ten clients from each parameter center in Table II. Mass and inertia receive independent uniform multiplicative scatter of $\pm 2.5\%$ and axle stiffnesses $\pm 5\%$. Geometry remains fixed. Family labels are withheld from primary block allocation and fitting.

Calibration follows curvature $\kappa_0 = 0.005$ m^{-1}. At each observed speed, the simulator sets $r_i^\star = v\kappa_0$ and solves for $\beta_i^\star, u_i^\star$ with $\dot{\beta} = \dot{r} = 0$, using that client's plant parameters. Each of 36 records starts independently from $x_i^\star + \Delta x_j$, holds steering $u_i^\star + \Delta u_j$ and speed fixed for one fourth-order Runge–Kutta (RK4) step of $T_s = 0.01$ s, and supplies

$$\begin{aligned} x_{j,+} &= \mathcal{F}_i^{\mathrm{RK4}}(x_i^\star + \Delta x_j, u_i^\star + \Delta u_j; v), \\ \widetilde{\Delta x}_{j,+} &= x_{j,+} - x_i^\star + \nu_j. \end{aligned} \tag{11}$$

The rows $[\Delta x_j^\top, \Delta u_j]$ and $\widetilde{\Delta x}_{j,+}^\top$ form Z and X_+ in (4); the three-column regressor is checked for full rank. These records are independently reset transitions, not consecutive driving samples. In the primary study, only next-state targets receive noise ν_j. Zero-mean independent Gaussian perturbation standard deviations are approximately 0.25° for β, 2°/s for r, and 0.30° for steering. Added next-state noise has standard deviations 0.01° and 0.03°/s.

A prospective workflow is instrumented commissioning: hold selected speeds near steady cornering, estimate state/steering centers from settled data, and apply small steering perturbations with synchronized speed, steering, yaw-rate, and sideslip measurements. Sideslip requires a reference sensor or validated estimator. Local regression pairs supply the shared statistics.

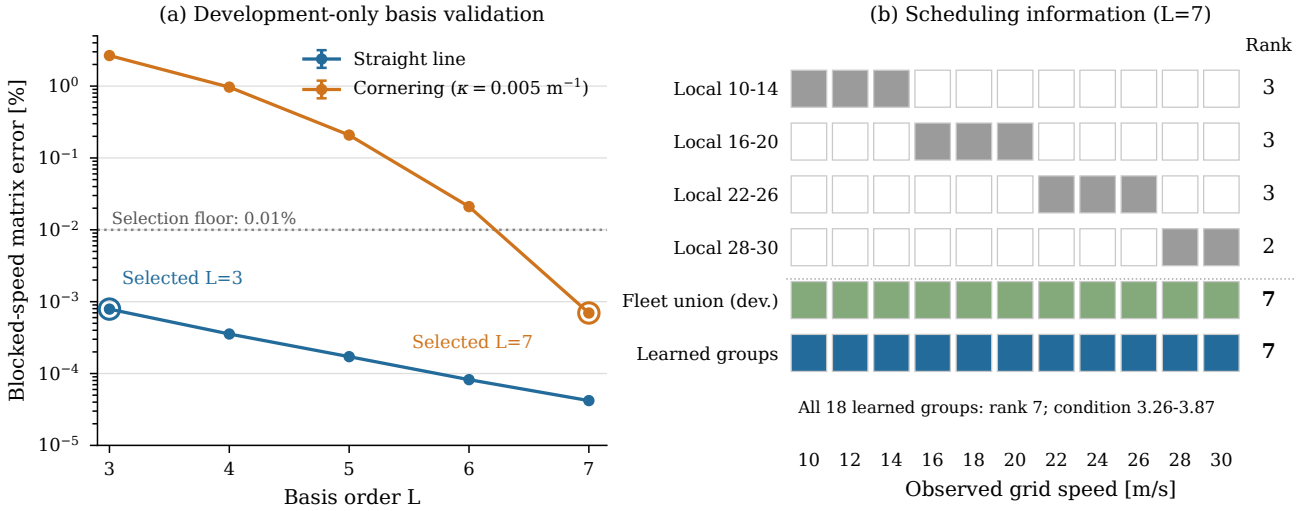


Fig. 2. Scheduling representation and information coverage. (a) Mean blocked-speed matrix error across ten development fleets; error bars denote one standard error. Rings mark the selected orders, and the dotted line is the 0.01% practical selection floor. (b) Filled cells mark observed grid speeds for the seven-function basis. The development union and all 18 audited primary groups have rank seven, unlike individual clients. Development union and learned-group coverage are distinct checks; full rank does not remove within-group bias.

This acquisition procedure remains untested: real maneuvers cannot reset states arbitrarily, and correlated regressors, achievable excitation, and center estimation require validation. The sensitivity tests perturb the reset-record benchmark, not sequential driving or observer errors.

The four local speed blocks are $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$, and $\{28, 30\}$ m/s. Globally balanced random allocation assigns them to eight, eight, seven, and seven clients, respectively, without consulting family labels. Primary fleet seeds are 371–380. Development representation selection uses separate seeds 301–310, contiguous blocked-speed validation, a one-standard-error rule, and a practical relative-error floor of 10^{-4}.

B. Endpoints, averaging, and baselines

Reference matrices $m_i^\star(v)$ are numerical continuous-time Jacobians at the prescribed equilibrium, discretized by an augmented matrix exponential. They serve development-only representation and penalty selection and evaluation, but do not enter finite-data group fitting. Full-envelope matrix error is evaluated on the common speed grid:

$$e_i(v) = \frac{\|\widehat{m}_i(v) - m_i^\star(v)\|_2}{\|m_i^\star(v)\|_2}, \qquad E_s = \frac{1}{30 \cdot 11} \sum_i \sum_{v \in \mathcal{V}_{\text{grid}}} e_i(v). \tag{12}$$

Here $\mathcal{V}_{\text{grid}} = \{10, 12, \ldots, 30\}$ m/s and s indexes fleets. The metric uses the stated physical coordinates without entry normalization. It is coordinate dependent and does not directly measure tracking performance. A post-hoc sensitivity check averages $\sqrt{\frac{1}{6}\sum_{j=1}^{6}[(\widehat{m}_{ij}(v) - m_{ij}^\star(v))/d_j]^2}$ over the same pairs. Each fixed d_j is the RMS reference entry across saved development fleets 321–330 and all eleven speeds; evaluation data do not select scales. This balances entry magnitudes without defining a control-weighted norm. Entry diagnostics use mean absolute error (MAE). Observed/unobserved-speed summaries average only their respective client-speed pairs within each fleet. Reported errors average E_s over ten independent fleets. For method M and baseline B, reported gains average $100(1 - E_s^M/E_s^B)$ over paired fleets, rather than taking the ratio of grand mean errors. Confidence intervals resample those ten fleet units 20,000 times.

Restricted Local uses $L = 3$, ridge 10^{-8}, and each client's records. Global and both grouped methods use $L = 7$ and the same fleet records. These baselines separate complementary coverage, compatibility selection, and distributed implementation. Prior-centered Local uses $L = 7$ with ridge toward its own $L = 3$ fit expressed in the richer basis, denoted $\Theta_{i,0}$. Leave-one-speed-out validation selects the penalty ρ_i from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}\}$; clients with only two speeds use the fixed middle candidate 10^{-2}.

To explain its near-equivalence to Restricted Local, define $R_i = Y_i - \Phi_i\Theta_{i,0}$, $\Delta_i = \widehat{\Theta}_i^{\mathrm{P}} - \Theta_{i,0}$, and $P_i = \Phi_i^\top\Phi_i + (\rho_i + \epsilon)I$. The centered-ridge normal equations imply

$$P_i\Delta_i = \Phi_i^\top R_i - \epsilon\Theta_{i,0}, \qquad N_i\Delta_i = -\frac{\epsilon}{\rho_i + \epsilon} N_i\Theta_{i,0}, \tag{13}$$

where N_i has orthonormal rows spanning $\ker\Phi_i$ and $\epsilon = 10^{-12}$. If $R_i = 0$ and $\epsilon = 0$, then $\Delta_i = 0$ for every $\rho_i > 0$. The three-function design interpolates two or three distinct-speed targets; ridge gives near-interpolation. The richer fit therefore retains the prior's unsupported directions apart from numerical shrinkage.

To test sharing against a distinct local prior, Smooth Local

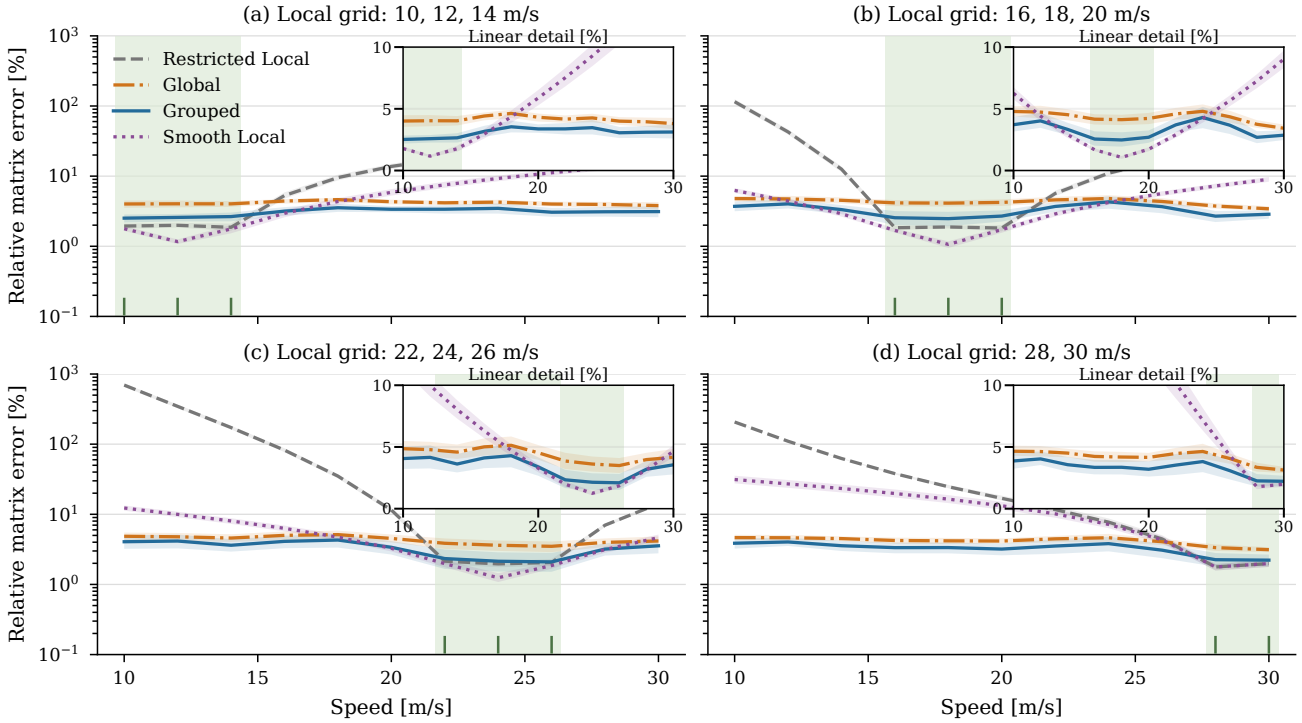


Fig. 3. Primary matrix error by local speed block, with family labels withheld. Light shading marks each observed block; short green marks identify its observed grid speeds. Curves average clients within block and fleet, then across ten fleets; bands are pointwise 95% fleet-bootstrap intervals from 20,000 draws. The logarithmic axis retains Restricted Local extrapolation failure. Smooth Local uses a development-selected curvature penalty and the same recipient targets. These curves and Fig. 4 summarize the primary cohort.

penalizes function curvature with $L = 7$:

$$\widehat{\Theta}_i^{\mathrm{S}} = \arg\min_{\Theta}\Big\{\frac{\|Y_i - \Phi_i\Theta\|_F^2}{n_i} + \alpha\|Q_2\Theta\|_F^2 + \eta\|\Theta\|_F^2\Big\}. \tag{14}$$

Here $\|Q_2\Theta\|_F^2$ uses 401-point trapezoidal quadrature for $\frac{1}{2}\int_{-1}^{1}\|d^2[\phi(v(z))^\top\Theta]/dz^2\|_2^2\,dz$; $\eta = 10^{-16}$ is numerical ridge. The curvature penalty leaves $\mathrm{span}\{1, z\}$ unpenalized, distinct from Restricted Local's $\mathrm{span}\{1, v^{-1}, v^{-2}\}$. Smoothness supplies a prior, without restoring local scheduling rank. A common α is selected from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2, 10^4, 10^6\}$ on development fleets 321–330 using unobserved-speed matrix error. The largest penalty within one fleet standard error of the best mean is chosen: 10^6, approaching an affine speed map. Reference Jacobians serve this offline tuning; recipient fitting uses only its own noisy targets at the original two or three speeds. This matched check and the group-information audit reuse the ten primary fleets and their saved calibration targets. They provide no independent confirmation. A full-coverage recipient reference outperforms family pooling at approximately equal transition budget, bounding the benefit to missing local access rather than superiority at equal recipient information.

C. Focused robustness protocol

To test the dependence on pooled coverage and calibration quality, ten fresh fleets (seeds 1201–1210) are evaluated under seven predeclared paired conditions. The reference repeats the primary procedure. All conditions retain each client's two- or three-speed count, 36 transitions per speed, physical parameters, basis, grouping settings, and the frozen Smooth Local penalty. Perturbation and noise draws are paired by client and speed-slot index; coverage changes alter speed locations while retaining record budgets.

For family-coverage restriction, all ten heavy clients observe the first two or three speeds in $\{10, 12, 14\}$ m/s; other clients retain their blocks. Labels enter only this stress generator and a separate OracleFamily diagnostic, never learned fitting or selection. Fleet-coverage restriction maps the four blocks to $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{10, 12, 14\}$, and $\{16, 18\}$ m/s, respectively. Four calibration conditions separately halve state/input excitation standard deviations, double next-state noise standard deviations, add current measurement noise with standard deviations $(0.01°, 0.03°/\mathrm{s}, 0.01°)$ for (β, r, u), or bias their assumed centers by magnitudes $(0.05°, 0.2°/\mathrm{s}, 0.1°)$. Bias signs are drawn independently per component/client and held fixed across its speeds. Center bias changes only the subtraction in (4), retaining identical measured records and zero-intercept OLS.

A predeclared rank-only fallback replaces deficient selected group maps with Global after selection, without changing grouping or fitting. Global deficiency is flagged separately. The fixed full-grid error (12) is the main endpoint; recipient-unobserved sets change with coverage. Intervals are pointwise 95% paired-fleet bootstrap intervals. No representation, prior,

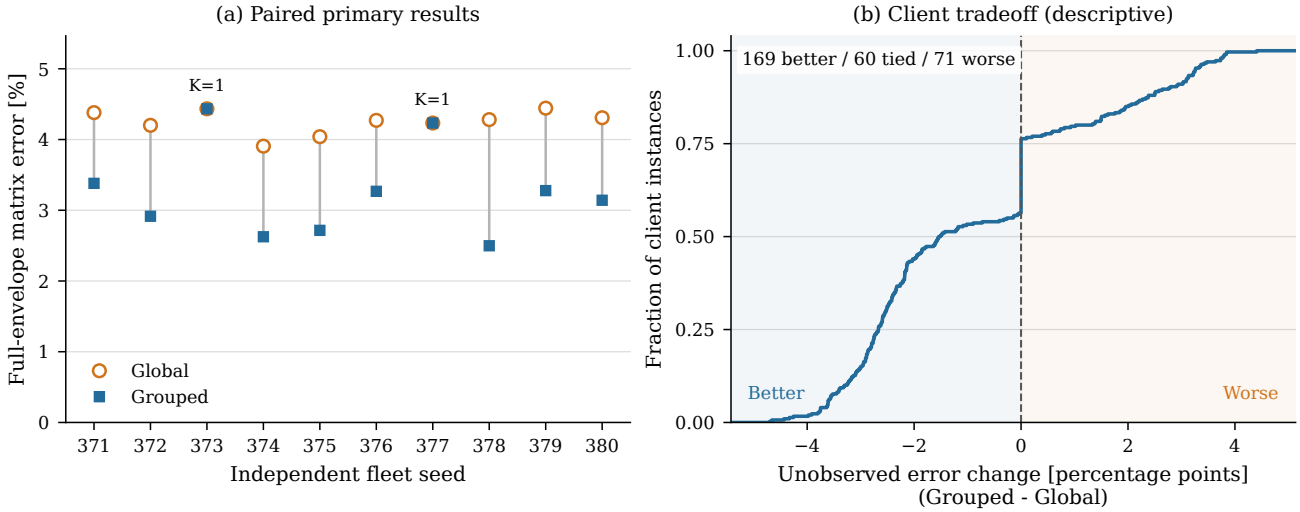


Fig. 4. Primary fleet gains and client tradeoffs. (a) Paired full-envelope errors for all ten independent fleets; connectors join methods within each fleet, and $K = 1$ marks the two ties. (b) Descriptive empirical cumulative distribution of client mean unobserved-speed error changes (Grouped minus Global) in percentage points; negative values indicate improvement. Ties use a 10^{-12} dimensionless-error tolerance. The 300 simulated clients are nested within ten fleets.

or controller is retuned, and no tracking rollout is added.

V. Identification Results

A. Why complementary coverage is needed

At curvature 0.005 m^{-1}, development blocked-speed matrix error falls from 2.6607% at $L = 3$ to 0.0007% at $L = 7$. Figure 2(a) contrasts this basis requirement with the straight-line case, which retains order three. For seven coefficients, the local blocks have ranks three, three, three, and two; the development union has rank seven and condition 3.406 (Fig. 2(b)). Complementary coverage can therefore supply missing directions, conditional on the learned partition.

With finite transitions and known compatibility, oracle-family pooling achieves 1.816% unobserved-speed error, versus 66.121% for Restricted Local and 4.296% for Global on separate development fleets. The prior-centered diagnostic changes Restricted Local negligibly there. These oracle results motivate the mechanism; the primary evaluation below withholds labels.

B. Primary label-free results

Table III separates the full-envelope endpoint from its observed and unobserved regions. Mean paired gains over Restricted Local and Global are 93.41% and 23.74%, respectively. The Global-relative 95% interval is $[15.39, 31.04]\%$, with improvement in eight of ten fleets. Two groups are selected in eight fleets and one in two. Recipient-only fits are more accurate at their own observed speeds, whereas sharing substantially reduces Restricted Local's unsupported extrapolation error. Grouped sharing additionally reduces Global's pooling error. Figure 3 locates these changes relative to each client block. Figure 4(a) shows the paired fleet errors, including the two $K = 1$ ties. Aggregate improvement does not extend to every client: 169 clients improve, 60 are numerical ties, and 71 have larger unobserved-speed error than Global. Figure 4(b) shows this tradeoff. Its 300 client clients are nested within ten fleets and are descriptive, rather than independent replicates for confidence intervals.

Prior-centered Local changes unobserved-speed error by only 0.00016 percentage points. Its prior's observed-target relative residual has median 0.0020% and maximum 0.0708% across 300 clients. Equation 13 explains the prior retention; its correction and nullspace identities are verified for all 300 saved fits.

Smooth Local reduces Restricted Local's extrapolation error and has the lowest observed-speed error (Table III). Against Smooth Local, grouped fitting gives mean paired gains of 56.20% in full-envelope error and 62.33% in unobserved-speed error. The 95% fleet-bootstrap intervals are $[49.69, 61.81]\%$ and $[57.03, 67.08]\%$, respectively. Both endpoints improve in all ten fleets. Figure 3 shows the remaining unobserved-speed gap. Other local priors remain untested.

The fixed-scale diagnostic retains mean paired gains over Global of 21.28% full-envelope and 17.93% unobserved, with pointwise fleet-bootstrap intervals $[13.65, 28.11]\%$ and $[10.94, 24.68]\%$. Entry-wise improvement is not uniform: unobserved B_1 MAE increases by 30.90% on average and deteriorates in every fleet; the A_{12} gain interval includes zero. This reuses primary fits and does not independently confirm them.

Federated and centralized protocols select identical partitions and group counts in all 20 fleet/protocol jobs, including the separate family-stratified coverage reference. Maximum coefficient discrepancy is 3.20×10^{-14} and maximum BIC discrepancy is 1.87×10^{-9}. This verifies the implementation under matched inputs.

C. Scheduling information in selected groups

All 18 primary selected groups contain observations at all eleven grid speeds and have scheduling-design rank seven. Their stacked design condition numbers range from 3.26 to

TABLE III
MEAN MATRIX ERROR [%] ACROSS TEN PRIMARY FLEETS WITH LABEL-FREE GROUPING. PRIOR-CENTERED LOCAL RETAINS THE RESTRICTED PRIOR; SMOOTH LOCAL TESTS FUNCTION CURVATURE.

Method	Full	Observed	Unobserved
Restricted Local	49.562	1.930	65.568
Prior-centered Local	49.562	1.930	65.568
Smooth Local	7.489	1.630	9.458
Global	4.250	3.840	4.388
Centralized groups	3.249	2.423	3.527
Federated groups	3.249	2.423	3.527

TABLE IV
ROBUSTNESS STUDY: MEAN FULL-GRID MATRIX ERROR [%] ACROSS TEN PAIRED FLEETS. n_1 IS THE NUMBER OF FLEETS SELECTING ONE GROUP ($K = 1$); THE OTHERS SELECT $K = 2$. SMOOTH DENOTES THE FROZEN SMOOTH LOCAL PRIOR.

Condition	Federated	Global	Smooth	n_1
Reference	3.058	4.214	7.221	3
Family restriction	3.879	4.584	7.779	5
Fleet restriction	33.423	33.255	6.645	5
Half excitation	4.324	4.324	13.179	10
Double target noise	4.326	4.326	13.286	10
Noisy regressors	4.039	4.258	9.707	7
Biased centers	4.810	4.906	9.734	9

3.87 in the fixed QR basis, as summarized in Fig. 2(b). The learned partitions satisfy Proposition 1's scheduling condition for this benchmark. The reported conditioning concerns the scheduling design, rather than its Gram or the full dynamical-identification information matrix. Full rank neither removes within-group bias nor guarantees physical-family recovery.

Retrospective family diagnostics clarify this distinction. In all eight $K = 2$ fleets, heavy and handling clients occupy separate groups, while nominal clients split between them. Labels remain withheld from learning. All 71 clients with larger unobserved error than Global are nominal; their family mean worsens in each $K = 2$ fleet. Thus compatible sharing is an aggregate regression compromise with sufficient coverage, rather than three-family recovery or uniform recipient benefit. These patterns do not establish why BIC selects two groups.

D. Coverage and calibration robustness

Table IV reports the robustness cohort separately from the primary fleets. Restricting heavy-family coverage leaves its ten-client OracleFamily design at rank three, despite exceeding the minimum size of five. Its family-specific full-grid error rises from 1.860% to 69.225%. Thus client count and known compatibility do not ensure scheduling information. The fleet and all selected learned groups remain rank seven. Each heavy-client group includes clients from other families. Federated fitting retains a mean paired 15.49% gain over Global, interval $[5.03, 26.76]\%$. This observed pooling response is not a rank guarantee from BIC.

Fleet-wide restriction instead gives rank six for Global and every selected or oracle-family group. Federated error reaches 33.423%, versus 6.645% for Smooth Local. On the five speeds absent from the entire fleet (22–30 m/s), their errors are 69.302% and 10.437%. The Global fallback is itself deficient and cannot supply the missing direction. Smoothness provides useful extrapolation in this condition, without restoring identifiability. An isolated deficient selected group with full-rank Global does not occur in this study; rescue of that case by the fallback is therefore not demonstrated.

Weaker excitation and doubled target noise select $K = 1$ in all ten fleets, making Federated identical to Global. Both reduce calibration signal-to-noise and give similar errors under the paired perturbations. Noisy regressors and biased centers also reduce the selected group count; their paired gain intervals over Global include zero. Sharing remains more accurate than Smooth Local in all four calibration conditions, but a grouping advantage over Global is not established there. All local transition designs retain rank three, and selected scheduling designs remain rank seven in these conditions: calibration sensitivity and scheduling deficiency are distinct limits. These are controlled identification tests, not online sensor-noise or tracking-robustness validation.

VI. SUPPORTING LQI VALIDATION

Ten separate confirmation fleets (seeds 411–420) refit the maps and apply speed-scheduled LQI to nonlinear 12 s maneuvers with available noise-free physical-state feedback. Each fleet repeats the partial-speed calibration above; maps remain fixed during tracking. ExactLPV is a recipient-specific $L = 7$ fit to exact Jacobians on the eleven-speed grid. At each grid node, the method's fitted $\widehat{A}, \widehat{B}$ define

$$A_{\mathrm{a}} = \begin{bmatrix} \widehat{A} & 0 \\ T_s c_r & 1 \end{bmatrix}, \quad B_{\mathrm{a}} = \begin{bmatrix} \widehat{B} \\ 0 \end{bmatrix}, \quad c_r = [0, 1]. \tag{15}$$

Discrete Riccati synthesis uses $Q = \mathrm{diag}(25, 50, 1000)$ for $[\beta, r, \xi]^\top$, with yaw-error integral ξ, and steering weight $R = 0.15$. The yaw-reference prefilter is $F = w + K_x h$, where $(I - \widehat{A})h - \widehat{B}w = 0$ and $c_r h = 1$; K_x denotes the first two entries of the LQI gain. For Restricted Local at unobserved grid nodes, synthesis evaluates its matrix map at the nearest observed speed. All methods then linearly interpolate gains and prefilters on the common grid. Weights were selected using only ExactLPV performance on development seeds 381–385, subject to feasibility and saturation checks. A weaker maneuver pilot on seeds 391–400 was discarded before the final confirmation. Neither grouped-versus-Global results nor confirmation fleets select weights. Smooth Local additionally reuses these fleets and calibration targets with frozen $\alpha = 10^6$, evaluating its map at all grid nodes. No weights, penalties, or fallbacks are retuned; this adds a comparator, not an independent cohort.

With $r_k^{\mathrm{ref}} = a_{y,k}^{\mathrm{req}}/v_k$, the runtime law is

$$\begin{aligned} u_k^{\circ} &= -K(v_k)[\beta_k, r_k, \xi_k]^\top + F(v_k) r_k^{\mathrm{ref}}, \\ \xi_{k+1}^{\mathrm{cand}} &= \mathrm{clip}_{[-0.6,0.6]}\{\xi_k + T_s(r_k - r_k^{\mathrm{ref}})\}. \end{aligned} \tag{16}$$

Steering is rate-limited relative to u_{k-1}, then amplitude-limited. The integral candidate is accepted when the command is effectively unsaturated or the integral increment counteracts

TABLE V

SUPPORTING YAW TRACKING RMSE [DEGREES/S], AVERAGED ACROSS TEN CONFIRMATION FLEETS. VIOLATIONS COUNT CLIENTS EXCEEDING THE 6° SIDESLIP BOUND (MODERATE/HARD), OUT OF 300 EACH. SMOOTH LOCAL MEANS INCLUDE FAILED RESPONSES. RESTRICTED LOCAL USES THE NEAREST OBSERVED SPEED OUTSIDE ITS BLOCK.

Method	Moderate	Hard	Violations
Restricted Local	0.0540	0.1698	0/0
Smooth Local	8.3565	5.7161	88/65
Global	0.0429	0.1564	0/0
Federated groups	0.0411	0.1546	0/0
ExactLPV reference	0.0400	0.1507	0/0

limiting; otherwise $\xi_{k+1} = \xi_k$. The integral bound is in radians. Each maneuver starts with $x_0 = 0$, $\xi_0 = 0$, and $u_{-1} = 0$; RK4 holds the applied steering and current speed fixed within each sample.

Feedback acts on absolute states, with reference feedforward from the homogeneous fitted matrices. Calibration equilibrium offsets are not used online. The maps describe one cornering curve, whereas the maneuvers vary the requested turn; tracking therefore tests empirical usability beyond that curve, rather than exact equilibrium-offset compensation.

The moderate maneuver smoothly varies speed $11 \to 29 \to 15$ m/s and uses an enveloped two-frequency lateral-acceleration demand with scale 4 m/s^2 and frequencies 0.35 and 0.72 Hz. The hard maneuver varies speed $29 \to 12 \to 28$ m/s and uses alternating Gaussian demands centered at $3.0, 6.0, 9.2$ s, with respective widths $0.40, 0.48, 0.55$ s and amplitudes $7, -7.7, 5.6$ m/s^2. Yaw reference is requested acceleration divided by speed. Steering limits are 12° and 120°/s, with a 6° sideslip feasibility bound. These remain fixed across methods. Yaw RMSE uses the complete maneuver, then averages clients within fleet and the ten fleet means.

For moderate and hard demands, mean paired tracking gains are 23.86% and 8.96% against Restricted Local, and 4.26% and 1.21% against Global, respectively. Global-relative intervals are $[1.95, 6.55]\%$ and $[0.47, 2.01]\%$. Absolute mean improvements are 0.001824 and 0.001874 degrees/s, so the control evidence supports usability with modest practical headroom. All federated trajectories satisfy the tested constraints. The maximum frozen-point spectral radius is 0.9759, evaluated on 81 speeds using the interpolated gain and ExactLPV reference augmentation (15). This unsaturated diagnostic on the calibration curve is not a scheduled or nonlinear stability certificate. Four fleets select one group, making federated and Global controllers identical there.

Smooth Local separates matrix accuracy from controller suitability. All 300 schedules synthesize and 600 rollouts finish numerically, but $114/300$ clients have frozen-point radius at least one (maximum 50.81). Sideslip violations affect $88/300$ moderate and $65/300$ hard runs. Severe tails dominate Table V; client-instance median RMSEs are 0.0589 and 0.2047 degrees/s. Excursions beyond the bound mark failed responses, not credible vehicle behavior. The tested prior does not reliably support this frozen synthesis protocol; other priors and separately tuned controllers remain untested.

Saved-map equations, schedules, and summaries were audited. A saved-schedule replay of fleet 411 agrees in all 60 feasibility flags but not trajectory hashes; individual RMSE differences reach 2.89 degrees/s. Table V retains executed results. Sensitive failed responses limit precise tail-error interpretation; feasibility is the main finding.

VII. DISCUSSION AND CONCLUSION

Complementary speed blocks supply missing local directions, while grouping reduces aggregate pooling error relative to Global. The primary advantage persists under development-scaled entry weighting, but does not extend to every entry or recipient. Heavy and handling clients benefit while many nominal clients deteriorate; learned groups need not recover physical families. Centralized agreement verifies the distributed implementation.

The robustness study bounds the benefit: minimum group size does not ensure rank, and Global fallback cannot restore fleet-wide missing information. Smooth Local wins in that deficient condition; weaker calibration signal-to-noise can collapse grouping to Global. Full selected-group rank is empirical, selection is heuristic, and exhaustive search incurs measurable communication. Ordinary driving, observer errors, real fleets, and frozen-library transfer remain unvalidated.

The scheduled-LQI tests support usability with modest gains over Global. The tested Smooth Local prior yields infeasible responses for some clients under the frozen protocol, illustrating that matrix accuracy alone does not establish controller suitability. Compatible sharing is useful when pooled coverage and calibration quality support it, with explicit recipient tradeoffs.

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